

Paper Review: Maximin

—A. Wall

Alvin Yang

1. Bibliographical information:

A. C. Wall, Maximin Surfaces, and the Strong Subadditivity of the Covariant Holographic Entanglement Entropy, *Class. Quant. Grav.* 31 (2014), no. 22 225007, [arXiv:1211.3494].

2. Main Claim:

The HRT surfaces that evolve (covariant) over time can be obtained through a construction called maximin. This kind of surfaces, on spacetimes with null curvature condition, has some properties, including the strong subadditivity mentioned in the title. And these surfaces exist at least in horizonless spacetimes and in black hole spacetimes with Kasner-like singularities. And these surfaces naturally have to be equivalent to the HRT surfaces.

3. Why is this interesting?:

The duality of Quantum Information and Geometry is one example of AdS/CFT correspondence. This is interesting because any correspondence are expected to have some implicit equivalence. And this conjectures that spacetime emerges from entanglement.

When relating entanglement entropy to geometry, the geometric dual must also satisfy strong subadditivity, a fundamental property of quantum entropy. However, during dynamical time evolution, the original proof formulated for static RT surfaces no longer applies. Dynamically evolving extremal (HRT) surfaces must also obey strong subadditivity: proving this is the central goal of the paper and it appears in the title. While HRT surfaces were proposed conceptually, a practical technical method to find them was lacking. Maximin provides precisely such a method, making it easier to demonstrate properties required by quantum information, such as strong subadditivity.

4. Guess how the result was obtained:

In fact, the author explicitly outlines the method to be used: minimizing the area on an achronal slice and then maximizing it with respect to variations of the slice over time.

Prior to reading this, I guessed the extremal surfaces would be obtained using variational methods to find extremum because “max” and “min” appear in the title. As for strong subadditivity, I guessed it would be derived by proving a duality between the quantum information side and the geometry side, showing that once strong subadditivity holds in quantum information theory, it must hold geometrically as well.

5. My guess vs. actual paper:

The actual paper takes a different route. The author aims to prove strong subadditivity directly on the geometric side, thereby strengthening confidence in the information-geometry duality itself. Regarding the method for obtaining maximin surfaces, the author explicitly defines the construction procedure. And the application of the variational method that I guessed in this article is to calculate the part of the RT surface, and this part is a part in the maximin surface's construction. The author further addresses the selection problem when multiple surfaces satisfy the criteria, leading to the introduction of the stability concept. To justify the validity of the maximin construction, the author proves its equivalence to HRT surfaces, and subsequently explores its various properties.

This updates my understanding of the paper: the author reformulates HRT surfaces in an equivalent but distinct way, whose chief advantage is simplifying the proofs of key properties. Once these geometric properties match expectations (aligning with entropy inequalities), they strengthen confidence in the validity of the HRT conjecture.

6. Regime of Validity:

Many conclusions proven in this paper require the Null Curvature Condition (NCC). The bulk spacetime is assumed to be classical, smooth, asymptotically locally AdS, AdS-hyperbolic, and compact after conformal compactification, as detailed in the “Assumptions about the Spacetime” section.

Furthermore, many results rely on the existence of maximin surfaces. The author proves their existence for horizonless spacetimes and black hole spacetimes with Kasner-like singularities, leaving other configurations unexplored. The author explicitly notes that spacetimes with global horizons, such as “bag of gold” geometries, are not considered in this paper.

Some of the less mathematically rigorous parts of this article are acceptable in practice because they can be handled by making certain and conventional physical assumptions, and thus are not needed to be discussed here.

7. What would you do next?:

Personally, a natural next step would be to extend these results to a broader class of spacetimes. It is possible that in certain geometries, these conclusions—or even the covariant holographic entanglement entropy conjecture itself—might face limitations, warranting further theoretical modifications.

The author addresses this in the Discussion section, suggesting that the maximin technique could yield additional results and that existence proofs could be extended, which aligns with my thought. The author also suggests pushing beyond the classical GR limit. Finally, formalizing the mathematics with greater rigor represents another feasible avenue of future work, particularly for mathematicians.

8. Mind Palace Entry:

By reformulating HRT surfaces through a maximin construction, this paper provides a geometrically convenient framework on spacetimes satisfying specific conditions (e.g., NCC, horizonless or Kasner-type) to easily prove properties like strong subadditivity, thereby illustrating the duality between quantum information and spacetime geometry and reinforcing the validity of the HRT conjecture.